Program No: Project	Date: 23-02-2026

Program Title: Predictive Modeling of Obesity Levels Based on Lifestyle and Dietary Habits: A Comparative Study using Orange Data Mining

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Roll no: Student 02

Obesity level analysis using orange

Objective of the Project

To analyze behavioral and physical data to predict a person's obesity category (e.g., Underweight, Normal Weight, Obesity Type I, II, or III) using classification models in Orange Data Mining. The goal is to identify which lifestyle factors—such as caloric intake, physical activity, and technology use—are the strongest predictors of weight-related health risks.

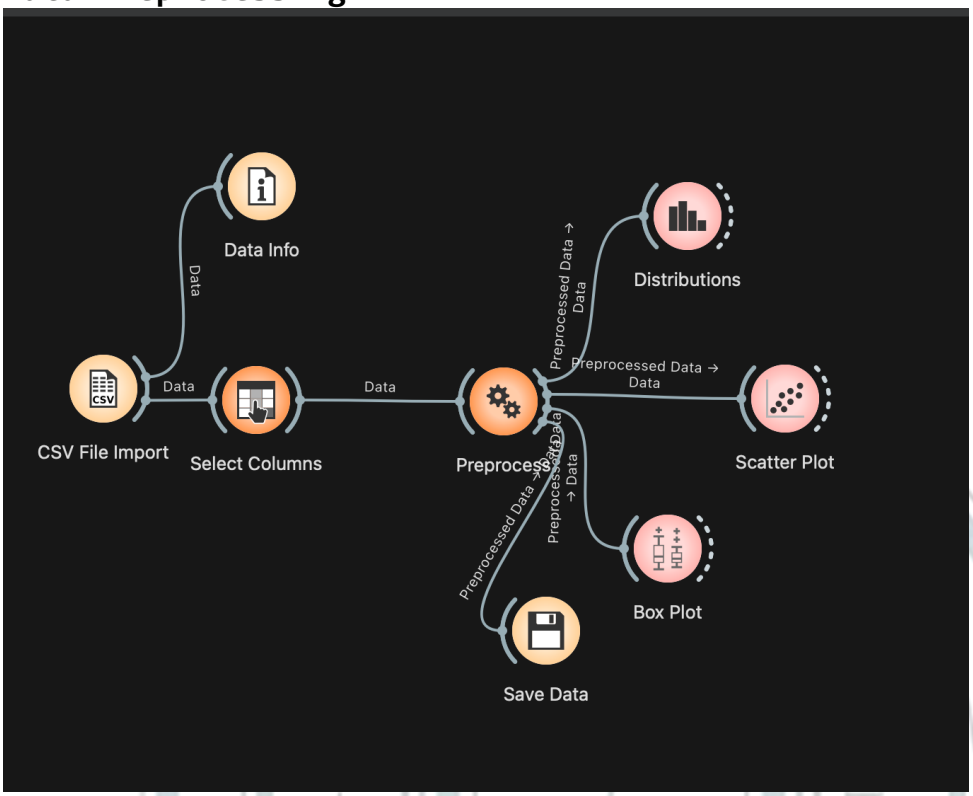
Dataset Description

- Dataset Name: Obesity Levels Based on Eating Habits and Physical Condition
- File Format: CSV (Comma-Separated Values)
- Total Instances: 1,689 (Training), 422 (Test) — *Note: This assumes an 80/20 split of the standard 2,111 row dataset.*
- Target Variable: NObeyesdad (Obesity Level)
- Attributes: Gender, Age, Height, Weight, Family History of Overweight, FAVC (High caloric food consumption), FCVC (Vegetable consumption), NCP (Number of main meals), CAEC (Consumption of food between meals), SMOKE, CH2O (Water intake), SCC (Calories monitoring), FAF (Physical activity frequency), TUE (Time using technology), CALC (Alcohol consumption), MTRANS (Transportation used).

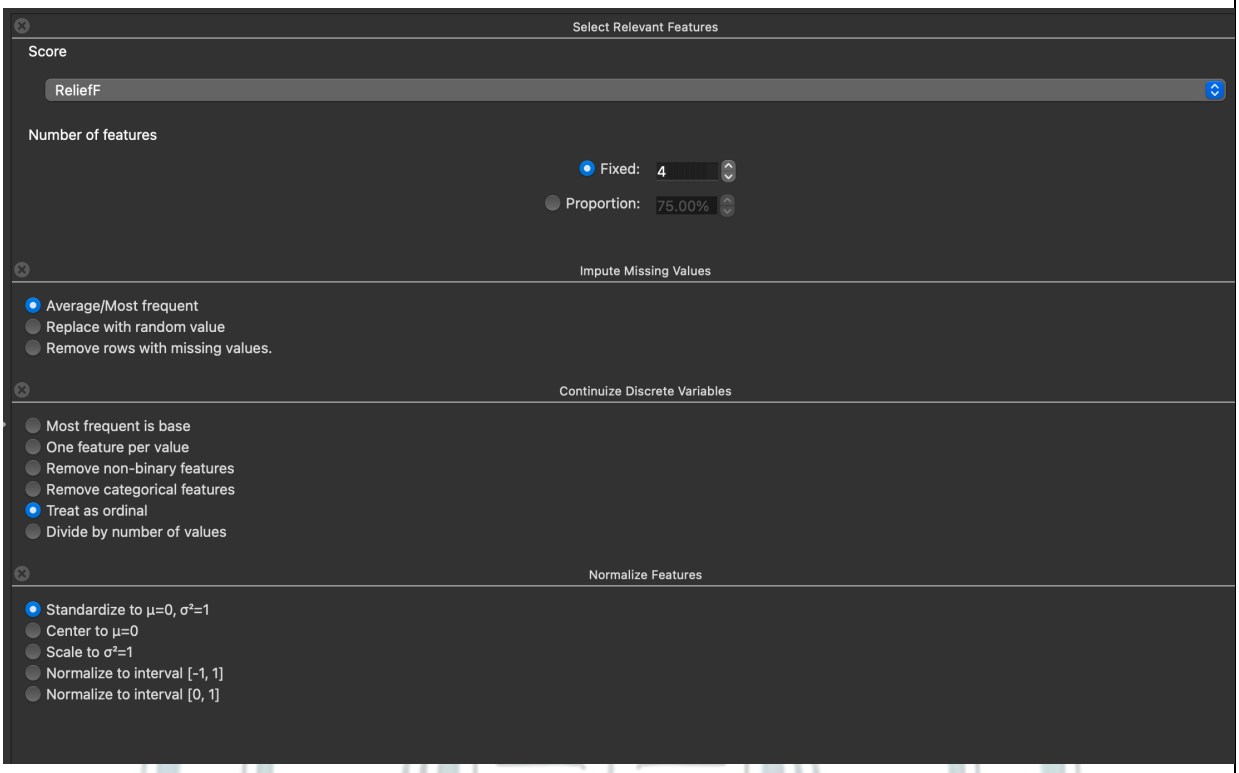
ObesityDataSet_raw_and_data_sinthetic

Age	Gender	Height	Weight	CALC	FAVC	FCVC	NCP	SCC	SMOKE	CH2O	family_history_with_overweight	FAF	TUE	CAEC	MTRANS	NObeyesdad
21	Female	1.62	64	no	no	2	3	no	no	2	yes	0	1	Sometimes	Public_Transportation	Normal_Weight
21	Female	1.52	56	Sometimes	no	3	3	yes	yes	3	yes	3	0	Sometimes	Public_Transportation	Normal_Weight
23	Male	1.8	77	Frequently	no	2	3	no	no	2	yes	2	1	Sometimes	Public_Transportation	Normal_Weight
27	Male	1.8	87	Frequently	no	3	3	no	no	2	no	2	0	Sometimes	Walking	Overweight_Level_
22	Male	1.78	89.8	Sometimes	no	2	1	no	no	2	no	0	0	Sometimes	Public_Transportation	Overweight_Level_
29	Male	1.62	53	Sometimes	yes	2	3	no	no	2	no	0	0	Sometimes	Automobile	Normal_Weight
23	Female	1.5	55	Sometimes	yes	3	3	no	no	2	yes	1	0	Sometimes	Motorbike	Normal_Weight
22	Male	1.64	53	Sometimes	no	2	3	no	no	2	no	3	0	Sometimes	Public_Transportation	Normal_Weight
24	Male	1.78	64	Frequently	yes	3	3	no	no	2	yes	1	1	Sometimes	Public_Transportation	Normal_Weight
22	Male	1.72	68	no	yes	2	3	no	no	2	yes	1	1	Sometimes	Public_Transportation	Normal_Weight
26	Male	1.85	105	Sometimes	yes	3	3	no	no	3	yes	2	2	Frequently	Public_Transportation	Obesity_Type_I
21	Female	1.72	80	Sometimes	yes	2	3	yes	no	2	yes	2	1	Frequently	Public_Transportation	Overweight_Level_
22	Male	1.65	56	Sometimes	no	3	3	no	no	3	no	2	0	Sometimes	Public_Transportation	Normal_Weight
41	Male	1.8	99	Frequently	yes	2	3	no	no	2	no	2	1	Sometimes	Automobile	Obesity_Type_I
23	Male	1.77	60	Sometimes	yes	3	1	no	no	1	yes	1	1	Sometimes	Public_Transportation	Normal_Weight
22	Female	1.7	66	Sometimes	no	3	3	yes	no	2	yes	2	1	Always	Public_Transportation	Normal_Weight
27	Male	1.93	102	Sometimes	yes	2	1	no	no	1	yes	1	0	Sometimes	Public_Transportation	Overweight_Level_
29	Female	1.53	78	no	yes	2	1	no	no	2	no	0	0	Sometimes	Automobile	Obesity_Type_I
30	Female	1.71	82	no	yes	3	4	no	yes	1	yes	0	0	Frequently	Automobile	Overweight_Level_
23	Female	1.65	70	Sometimes	no	2	1	no	no	2	yes	0	0	Sometimes	Public_Transportation	Overweight_Level_
22	Male	1.65	80	no	no	2	3	no	no	2	yes	3	2	Sometimes	Walking	Overweight_Level_
52	Female	1.69	87	no	yes	3	1	no	yes	2	yes	0	0	Sometimes	Automobile	Obesity_Type_I
22	Female	1.65	60	Sometimes	yes	3	3	no	no	2	yes	1	0	Sometimes	Automobile	Normal_Weight
22	Female	1.6	82	Sometimes	yes	1	1	no	no	2	yes	0	2	Sometimes	Public_Transportation	Obesity_Type_I
21	Male	1.85	68	Sometimes	yes	2	3	no	no	2	yes	0	1	Sometimes	Public_Transportation	Normal_Weight
20	Male	1.6	50	no	no	2	4	no	yes	2	yes	3	2	Frequently	Public_Transportation	Normal_Weight
21	Male	1.7	65	Always	yes	2	1	no	no	2	yes	1	2	Frequently	Walking	Normal_Weight
23	Female	1.6	52	Sometimes	yes	2	4	no	no	2	no	2	1	Frequently	Automobile	Normal_Weight

Data Preprocessing



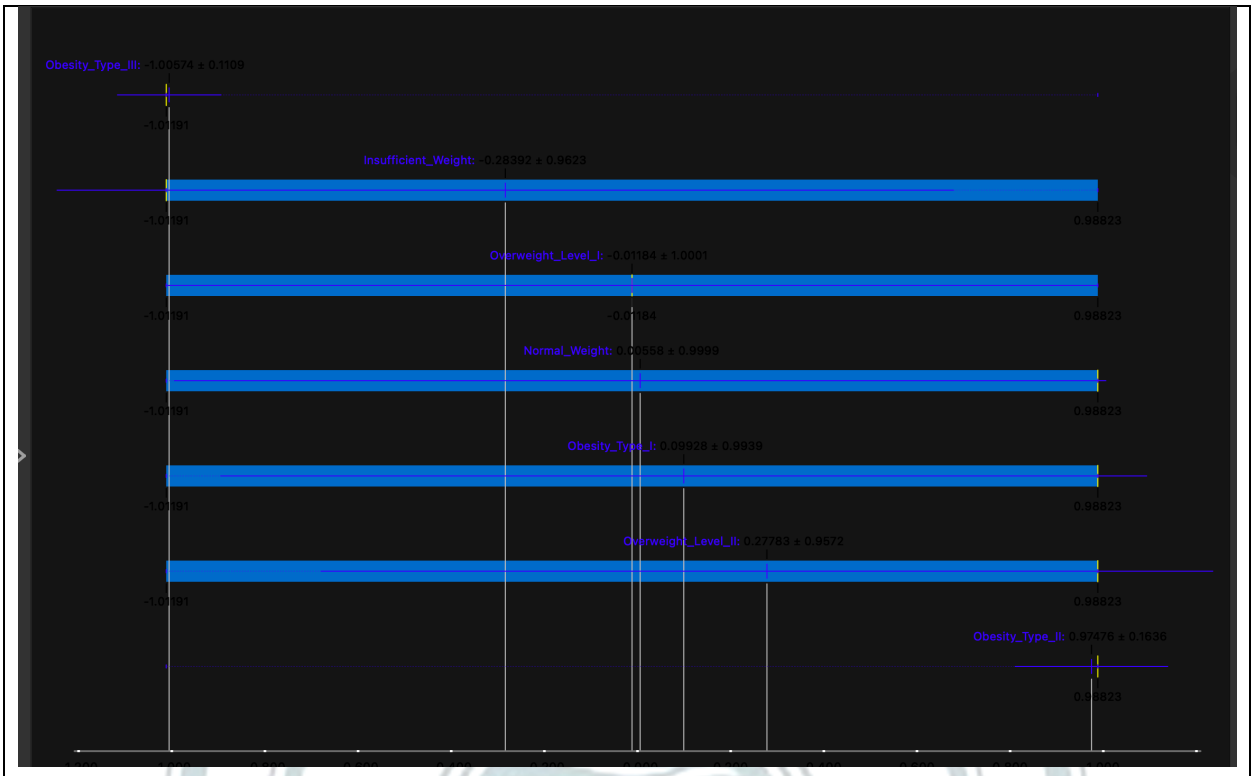
- **Data Loading:** Loaded the **Obesity Levels** dataset using the **File** widget in Orange.
- **Target Variable Selection:** In the **Edit Domain** widget, designated **NObeyesdad** (Obesity Level) as the target variable for classification.
- **Data Quality Check:** Verified the dataset for missing values and handled any inconsistencies to ensure the reliability of the feature set.
- **Preprocessing & Encoding:** Categorical attributes such as "Family History of Overweight," "Consumption of food between meals (CAEC)," and "Gender" were automatically encoded into numeric values using the **Preprocess** widget to make them compatible with machine learning algorithms.
- **Feature Normalization:** Applied **Standardization (z-score)** to numeric attributes like **Age**, **Height**, and **Weight**. This ensures that variables with larger numerical ranges do not dominate the model training process compared to smaller-scale variables.



Data Visualization

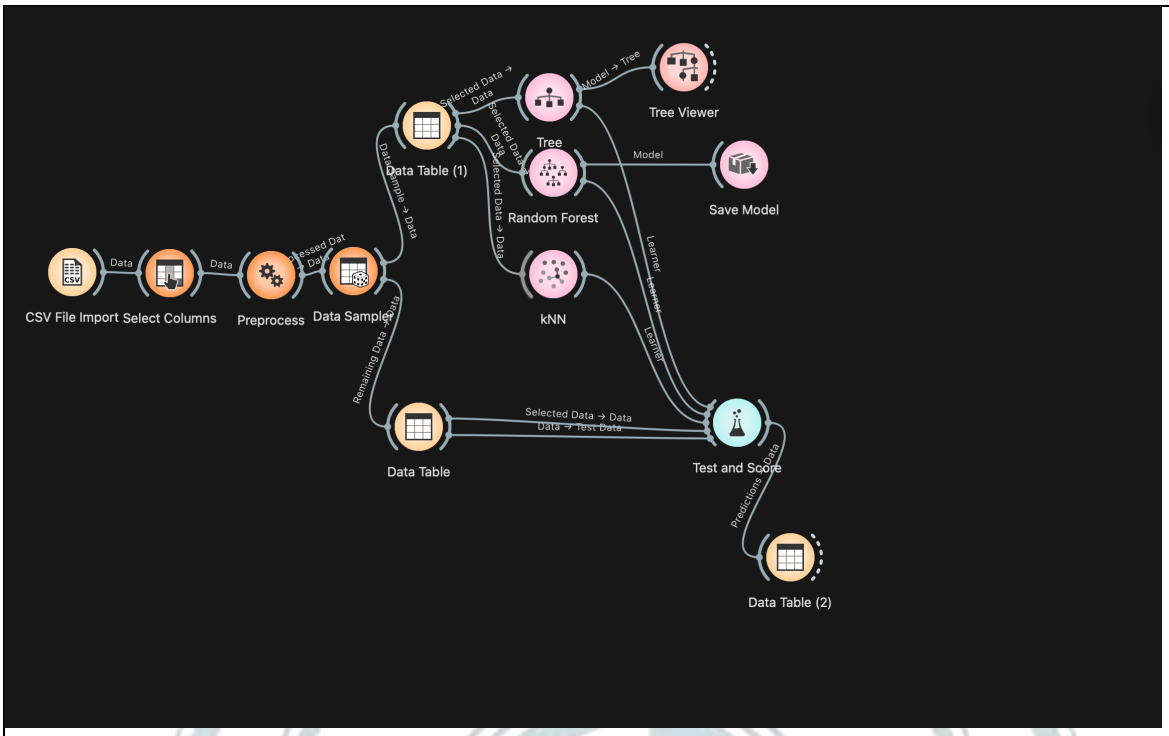
- Used Box Plot to analyze the distribution of Weight (or BMI) across different Obesity Levels (NObeyesdad). This helped identify the median weight and variance for each health category.

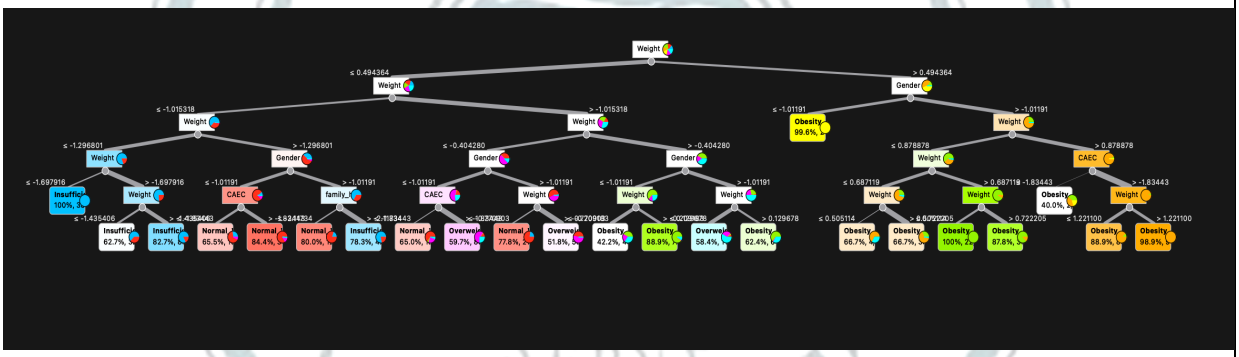
- **Used Scatter Plot** to visualize the relationship between **Age and BMI**. This allowed for the observation of how weight categories cluster and whether specific age groups are more prone to higher obesity levels.
- **Used Distributions** to observe the frequencies of key lifestyle and dietary attributes, such as **Physical Activity Frequency (FAF)** and **High Caloric Food Consumption (FAVC)**, to understand the general habits of the sampled population.



Model Building

- Split the dataset into training and testing sets using Data Sampler.
- Built the following classification:-Decision Tree,KNN,Random Forest


CSV File Import
Data
Select Columns
Data
Preprocess
Preprocessed Data
Data Sampler
Data Table (1)
Data Sample → Data
Remaining Data → Data
Selected Data → Data
Selected Data → Data
Selected Data → Data
Tree
Model → Tree
Tree Viewer
Model
Save Model
Random Forest
kNN
Learner
Learner
Learner
Data Table
Selected Data → Data
Data → Test Data
Test and Score
Predictions → Data
Data Table (2)



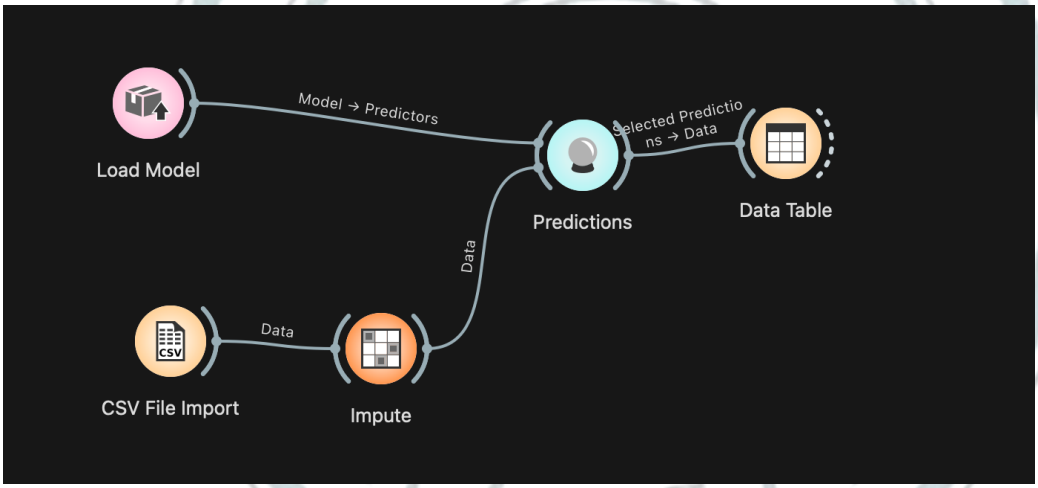
Model Evaluation

- Evaluated all models using the Test & Score widget.
- Compared models based on accuracy and other performance metrics.
- Random Forest was identified as the best-performing model.

Evaluation results for target (None, show average over classes)

Model	AUC	CA	F1	Prec	Recall	MCC
Tree	0.934	0.711	0.701	0.707	0.711	0.663
Random Forest	0.943	0.744	0.743	0.744	0.744	0.700
kNN	0.928	0.746	0.742	0.743	0.746	0.703

Prediction



- Saved the best model (Random Forest) using the Save Model widget.
- Loaded the model using Load Model.
- Applied the model to an unknown dataset.
- Viewed predicted results using the Predictions and Data Table
- widgets.

	Random Forest	LC	FAVC	FCVC	NCP	SCC	SMOKE	CH2O	history_with_over	FAF	TUE	CAEC	MTRANS	NObeyesdad
1	Overweight_Level_II		no	2	3	no	no	2	yes	0	1	Sometimes	Public_Trans...	Normal_Wei...
2	Overweight_Level_II	nes	no	3	3	yes	yes	3	yes	3	0	Sometimes	Public_Trans...	Normal_Wei...
3	Overweight_Level_II	ntly	no	2	3	no	no	2	yes	2	1	Sometimes	Public_Trans...	Normal_Wei...
4	Overweight_Level_II	ntly	no	3	3	no	no	2	no	2	0	Sometimes	Walking	Overweight_...
5	Overweight_Level_II	nes	no	2	1	no	no	2	no	0	0	Sometimes	Public_Trans...	Overweight_...
6	Overweight_Level_II	nes	yes	2	3	no	no	2	no	0	0	Sometimes	Automobile	Normal_Wei...
7	Overweight_Level_II	nes	yes	3	3	no	no	2	yes	1	0	Sometimes	Motorbike	Normal_Wei...
8	Overweight_Level_II	nes	no	2	3	no	no	2	no	3	0	Sometimes	Public_Trans...	Normal_Wei...
9	Overweight_Level_II	ntly	yes	3	3	no	no	2	yes	1	1	Sometimes	Public_Trans...	Normal_Wei...
10	Overweight_Level_II		yes	2	3	no	no	2	yes	1	1	Sometimes	Public_Trans...	Normal_Wei...
11	Overweight_Level_II	nes	yes	3	3	no	no	3	yes	2	2	Frequently	Public_Trans...	Obesity_Typ...
12	Overweight_Level_II	nes	yes	2	3	yes	no	2	yes	2	1	Frequently	Public_Trans...	Overweight_...
13	Overweight_Level_II	nes	no	3	3	no	no	3	no	2	0	Sometimes	Public_Trans...	Normal_Wei...
14	Overweight_Level_II	ntly	yes	2	3	no	no	2	no	2	1	Sometimes	Automobile	Obesity_Typ...
15	Overweight_Level_II	nes	yes	3	1	no	no	1	yes	1	1	Sometimes	Public_Trans...	Normal_Wei...
16	Overweight_Level_II	nes	no	3	3	yes	no	2	yes	2	1	Always	Public_Trans...	Normal_Wei...
17	Overweight_Level_II	nes	yes	2	1	no	no	1	yes	1	0	Sometimes	Public_Trans...	Overweight_...
18	Overweight_Level_II		yes	2	1	no	no	2	no	0	0	Sometimes	Automobile	Obesity_Typ...
19	Overweight_Level_II		yes	3	4	no	yes	1	yes	0	0	Frequently	Automobile	Overweight_...
20	Overweight_Level_II	nes	no	2	1	no	no	2	yes	0	0	Sometimes	Public_Trans...	Overweight_...
21	Overweight_Level_II		no	2	3	no	no	2	yes	3	2	Sometimes	Walking	Overweight_...
22	Overweight_Level_II		yes	3	1	no	yes	2	yes	0	0	Sometimes	Automobile	Obesity_Typ...
23	Overweight_Level_II	nes	yes	3	3	no	no	2	yes	1	0	Sometimes	Automobile	Normal_Wei...
24	Overweight_Level_II	nes	yes	1	1	no	no	2	yes	0	2	Sometimes	Public_Trans...	Obesity_Typ...
25	Overweight_Level_II	nes	yes	2	3	no	no	2	yes	0	1	Sometimes	Public_Trans...	Normal_Wei...
26	Overweight_Level_II		no	2	4	no	yes	2	yes	3	2	Frequently	Public_Trans...	Normal_Wei...
27	Overweight_Level_II		yes	2	1	no	no	2	yes	1	2	Frequently	Walking	Normal_Wei...
28	Overweight_Level_II	nes	yes	2	4	no	no	2	no	2	1	Frequently	Automobile	Normal_Wei...